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Program/Section: IT-IT21S6	Instructor: Mr. Paul Michael C. Arevalo
Activity Title: Final Project Guide OOP	

Project Title: Object-Oriented Hotel Reservation System

Course: Object Oriented Programming
Deliverable: Functional Hotel Reservation Application + Room & Reservation Specifications + Technical Documentation

Key Requirements & Mathematical Specs

Hotel Reservation Application

- **Number of Nights:** Number of Nights = Check-out Date − Check-in Date
- **Room Cost:** Room Cost = Room Rate × Number of Nights
- **Discount:** Discount Amount = Room Cost × Discount Rate
- **Tax:** Tax = (Room Cost − Discount Amount) × Tax Rate
- **Final Bill:** Total = Room Cost − Discount Amount + Tax
- **Change:** Change = Cash Tendered − Total

Room & Reservation Specifications:

- **Room Details:**
    - Room Number
    - Floor
    - Room Type
    - Capacity
    - Description
    - Price per Day
    - Availability Status
    - Housekeeping Status

- **Room Management:**
    - Add new rooms.
    - Update existing room information.
    - Update room type, capacity, description, and price.
    - Search and display room information.
    - Check room availability.
    - Update room occupancy status.

- **Housekeeping Status:** Manually updating and adjusting the room's operational status
    - Clean
    - Dirty
    - Under Maintenance
    - Out of Service

- **Room Monitoring:** A dashboard showing occupancy status, total available rooms per type, and clean-up or maintenance alerts.
    - Available rooms
    - Occupied rooms
    - Rooms by room type
    - Rooms requiring cleaning
    - Rooms under maintenance
    - Rooms out of service

OOP Classes:

- **Rooms:** Encapsulates room features, rates, availability status, housekeeping status, and methods for updating room information.
    - Room Number
    - Floor
    - Room Type
    - Capacity
    - Description
    - Price per Day
    - Availability Status
    - Housekeeping Status

- **Guest:** (Abstract Class): Defines the programmatic contract for calculating bills based on classification.
    - Full Name
    - Contact Number
    - Email
    - Guest Classification
    - Discount Eligibility/ID, if applicable

    - RegularGuest: Implements the 12% tax addition logic.
        - 12% tax
        - No discount
    - DiscountedGuest: Implements the 20% discount + tax exemption logic.
        - 20% discount
        - No tax

- **Room Repository / Inventory:** Manages room records, including adding new rooms, updating room details, checking room availability, adjusting housekeeping status, and displaying room information.
    - Adding rooms
    - Updating rooms
    - Searching rooms
    - Checking availability
    - Updating occupancy
    - Updating housekeeping status
    - Retrieving room information

- **Reservation:** Manages guest room reservations, including guest and room information, reservation dates, reservation status, room availability verification, reservation confirmation, and cancellation.
    - Guest information/reference
    - Room information/reference
    - Check-in Date
    - Check-out Date
    - Number of Nights
    - Reservation Type
        - Online
        - Walk-in
    - Reservation Status
        - Pending
        - Confirmed
        - Checked-In
        - Checked-Out
        - Cancelled
    - Reservation confirmation
    - Cancellation, if included

- **Transaction:** Processes check-out billing and payment, calculates the room charges based on the guest's stay and classification, calculates change, and generates the guest receipt.
    - **Reservation reference**
    - **Guest information/reference**
    - **Room information/reference**
    - **Number of Nights**
    - **Room Rate**
    - **Room Cost**
    - **Discount Amount**

- **Tax Amount**
- **Total Amount**
- **Cash Tendered**
- **Change**
- **Payment Status**
- **Receipt Generation**

Project Implementation Guide

1. System Design, Database & Room Management Module - Deadline: Week 1

Define the core OOP classes and database structure, then implement the basic room management functions for storing and managing hotel room information.

- Design and create the required database tables for rooms, guests, reservations, and transactions.
- Establish the Java-to-database connection.
- Design and create the required database tables.
- Implement adding new rooms and storing room details.
- Store the following room information:
  - Room Number
  - Floor
  - Room Type
  - Capacity
  - Description
  - Price per Day
  - Availability Status
  - Housekeeping Status
- Implement updating existing room information.
- Implement updating room type, capacity, description, and price.
- Implement room searching.
- Implement displaying room information.
- Implement room availability checking.
- Implement input validation for room information.
- Implement the Room Inventory/Repository for managing room records.

2. Guest Classification, Discount & Room Availability Module — Deadline: Week 2

Implement the guest class hierarchy and room availability logic using inheritance and polymorphism.

- Implement the **Guest abstract class/interface**.
- Implement guest information:
    - Full Name
    - Contact Number
    - Email
    - Guest Classification
    - Discount Eligibility/ID, if applicable
- Create the **RegularGuest** class.
- Create the **DiscountedGuest** class.
- Implement Regular Guest billing rules:
    - No discount
    - 12% tax
- Implement Discounted Guest billing rules:
    - 20% discount
    - Tax exemption
- Implement the verification/recording of Discount Eligibility/ID when applicable.
- Implement room availability checking.
- Implement housekeeping status:
    - Clean
    - Dirty
    - Under Maintenance
    - Out of Service
- Prevent reservations from being made for unavailable rooms.
- Connect guest and room information with the database.
- Implement methods for updating room occupancy and housekeeping status.

3. Reservation & Check-In/Check-Out Integration — Deadline: Week 3

Build the main reservation functionality by connecting the guest, room, and database modules.

- Create the **Reservation** class.
- Record the **reservation** type (Online or Walk-in).
- Implement guest and room selection.
- Record guest information for each reservation.
- Record the selected room.
- Record check-in date.
- Record check-out date.
- Calculate the number of nights:

Number of Nights = Check-out Date − Check-in Date

- Verify room availability before confirming a reservation.
- Prevent reservations for rooms that are:
    - Occupied
    - Under Maintenance
    - Out of Service
    - Otherwise unavailable
- Save reservation information to the database.
- Implement reservation confirmation.
- Implement check-in processing.
- Update the room's occupancy status after check-in.
- Implement check-out processing.
- Return the room to an available state after check-out, subject to housekeeping status.
- Update the room's housekeeping status when necessary.
- Implement reservation cancellation if included in the final system.

4. Billing, Payment & Receipt Module — Deadline: Week 4

Implement the reservation billing and payment process based **only on the room reservation**.

There will be **no additional room service charges** in the system.

- Calculate the number of nights.
- Calculate the room cost:

    Room Cost = Room Rate × Number of Nights

- Determine the guest classification.
- Apply the appropriate discount:

    Discount Amount = Room Cost × Discount Rate

- Calculate the applicable tax:

    Tax = (Room Cost − Discount Amount) × Tax Rate

- Calculate the final bill:

    Total = Room Cost − Discount Amount + Tax

- Process cash payments.

- Calculate the change:

  Change = Cash Tendered − Total

- Prevent the transaction when the cash payment is insufficient.
- Generate a guest receipt.
- Display the following information on the receipt:
  - Guest Name
  - Guest Classification
  - Room Number
  - Room Type
  - Check-in Date
  - Check-out Date
  - Number of Nights
  - Room Rate
  - Room Cost
  - Discount
  - Tax
  - Final Total
  - Cash Tendered
  - Change

5. Room Monitoring, System Integration & Testing — Deadline: Week 5

Integrate all system modules and implement the room monitoring dashboard before conducting final testing.

- Implement the **Room Monitoring Dashboard**.
- Display available rooms.
- Display occupied rooms.
- Display available rooms by room type.
- Display rooms requiring cleaning.
- Display rooms under maintenance.
- Display rooms that are out of service.
- Verify that room availability updates correctly after reservations.
- Verify that room occupancy updates correctly after check-in.
- Verify that room availability updates correctly after check-out.
- Verify that housekeeping status updates correctly.
- Test database operations and data consistency.
- Test Regular Guest billing calculations.
- Test Discounted Guest billing calculations.
- Test discount and tax calculations.
- Test payment and change calculations.

- Test invalid guest information.
- Test invalid room information.
- Test invalid dates.
- Test reservations for unavailable rooms.
- Test reservation cancellation if implemented.
- Fix errors and inconsistencies between modules.
- Polish the NetBeans interface and system output.
- Conduct final integration and system testing.
- Prepare the application and documentation for demonstration.